

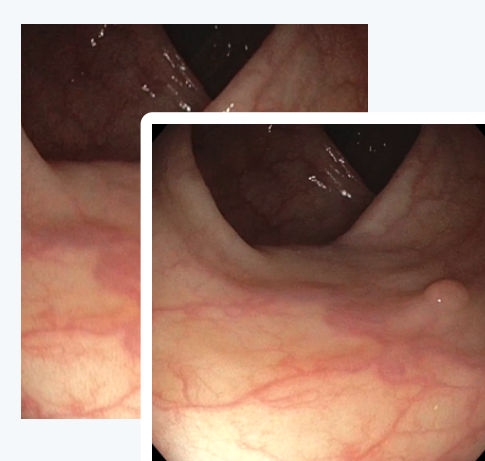
## Problem & Motivation

Colonoscopy produces large amounts of unlabelled video, while expert annotation is expensive and time-consuming. Useful representations must retain global meaning and small, transient findings despite substantial appearance variation.

### Joint-embedding methods

e.g. contrastive learning

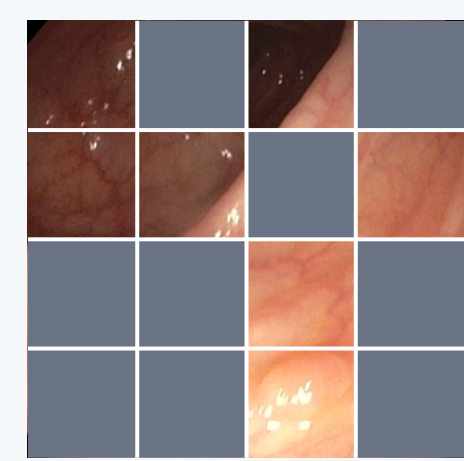
- Strong global semantics
- Can suppress fine detail through strong invariance



### Masked modelling methods

e.g. masked autoencoders

- Encourages local learning
- Pixel targets can favour appearance over semantics



### Endoscopy challenges

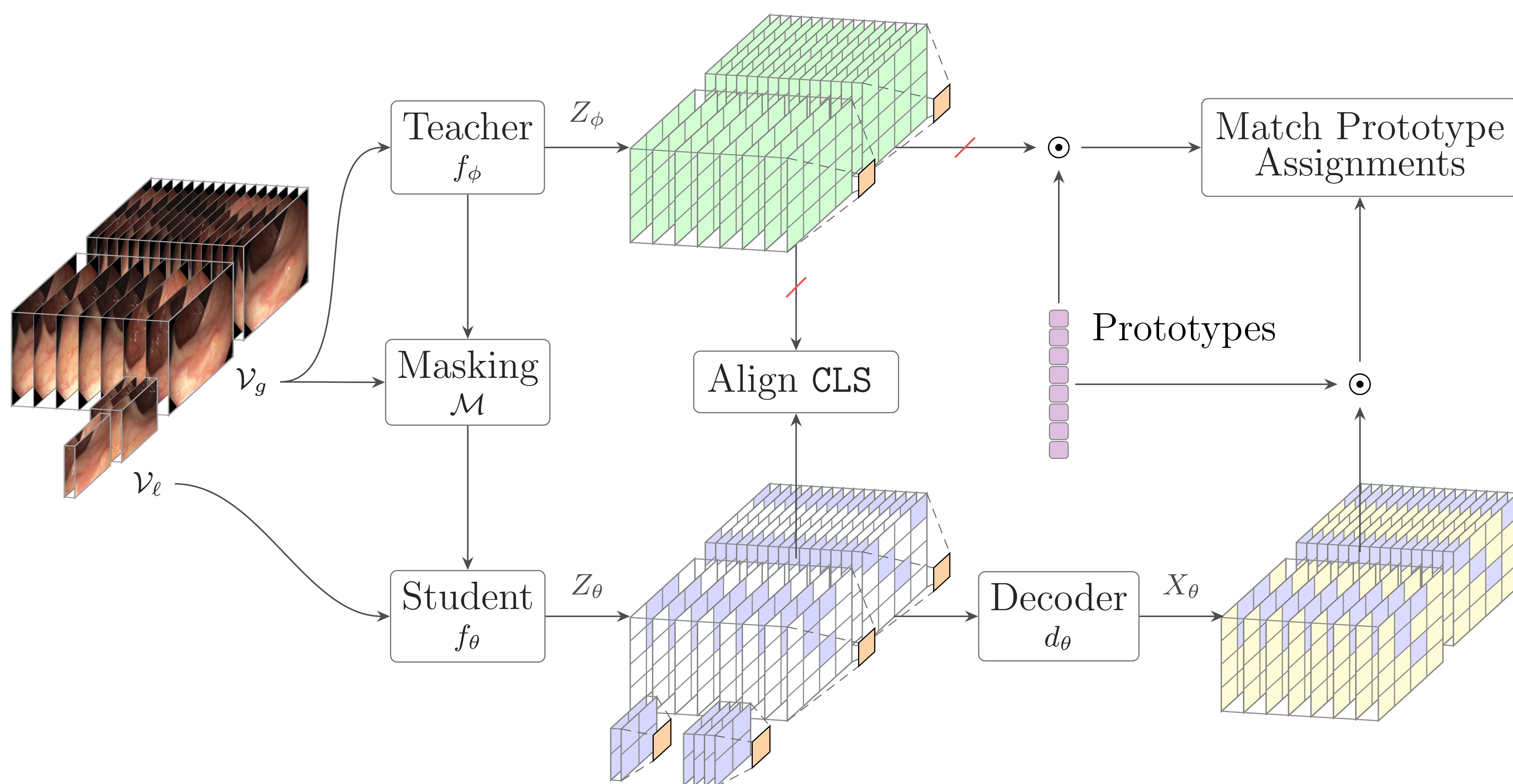


- Small, transient findings
- Substantial appearance variation
- Video token counts make pre-training expensive

**Goal: Learn representations that capture global semantics and fine-grained local detail, while keeping video pre-training computationally efficient.**

## STOMP - Semantic Token Masked Pre-training

Solution: A shared teacher guides global alignment and local semantic prediction from sparse student inputs.



Green: teacher features   Purple: student features   Yellow: predicted features   Orange: CLS token   Red slash: stop-gradient

### 1. Encode efficiently

The teacher sees complete global clips. The student drops 80% of global patch tokens before encoding and also uses local crops.

### 2. Align global meaning

Align CLS tokens across different views of the same clip. Local crops contribute to this global consistency objective.

### 3. Predict local semantics

Predict features at masked global locations. Match teacher and student at the same sites using balanced targets over 1,024 prototypes.

## Downstream performance

| Method             | PolypDiag<br>Classification F1 | CVC-12k<br>Segmentation Dice | KUMC<br>Detection F1 |
|--------------------|--------------------------------|------------------------------|----------------------|
| M <sup>2</sup> CRL | 94.2 ± 0.7                     | 81.4 ± 0.8                   | 86.3 ± 0.8           |
| STOMP              | 96.5 ± 0.8                     | 84.0 ± 0.2                   | 86.1 ± 1.1           |

Mean ± SD over three seeds

## Training cost per step

### Step time

|                    |      |   |
|--------------------|------|---|
| Endo-FM            | 1.38 | s |
| M <sup>2</sup> CRL | 2.29 | s |
| STOMP              | 0.70 | s |

### Peak allocated GPU memory

|                    |      |     |
|--------------------|------|-----|
| Endo-FM            | 35.8 | GiB |
| M <sup>2</sup> CRL | 67.7 | GiB |
| STOMP              | 20.6 | GiB |

## Conclusion

STOMP shows that local semantic supervision and global consistency can coexist with sparse encoding. The complementary objectives balance transfer across tasks while reducing pre-training cost. Lower cost may support longer clips, higher resolutions and larger endoscopy datasets.

Paper  
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